

RESEARCH

Open Access



Cancer pain and quality of life structural equation model based on the biopsychosocial model

Xiaoyu Hu^{1,5}, Jingjing Lü^{2,3}, Chang Qin^{1,4}, Xiaoxuan Liu^{1,5}, Jianbiao Zhu^{1,4} and Yongchun Cui^{1,5*}

Abstract

Objective This study employed a biopsychosocial framework to develop structural equation models (SEM) quantifying the multidimensional impact of pain on quality of life in cancer patients, aiming to provide scientific evidence and novel clinical perspectives for evidence-based pain management strategies.

Methods This cross-sectional study recruited participants experiencing chronic cancer-related pain using convenience sampling from a tertiary cancer hospital in Shandong Province, China, between January and February 2024. Data collection involved face-to-face administration of the Brief Pain Inventory-Short Form and the Quality of Life Questionnaire-Function 17. Descriptive statistics and reliability analyses (Cronbach's α) were performed using SPSS 27.0. Confirmatory factor analysis and SEM were subsequently conducted using AMOS 23.0, with parameters estimated via the maximum likelihood method.

Results Two SEMs were developed to elucidate relationships between cancer pain and quality of life. The first-phase SEM demonstrated acceptable fit ($\chi^2/df = 2.653$, GFI = 0.826, RMSEA = 0.078, TLI = 0.902, CFI = 0.916), with all indices meeting established thresholds without modification. In this model, pain intensity significantly impaired cognitive function ($B = 0.165$, $P < 0.001$) but showed no significant associations with other quality-of-life domains. Pain interference with activity substantially compromised physical ($B = 0.576$), role ($B = 0.585$), cognitive ($B = 0.273$), and social functioning ($B = 0.260$), all $P < 0.001$. While pain interference with affectivity improved physical ($B = -0.467$) and role functioning ($B = -0.455$), both $P < 0.001$. The refined second-phase SEM exhibited improved fit ($\chi^2/df = 2.691$, GFI = 0.826, RMSEA = 0.078, TLI = 0.900, CFI = 0.912) after modification. In this model, pain significantly impaired cross-domain quality of life: physical function ($B = 0.480$), role function ($B = 0.540$), emotional function ($B = 0.195$), cognitive function ($B = 0.715$), and social function ($B = 0.369$), all $P < 0.001$.

Conclusion Pain has a negative impact on all aspects of quality of life, and different dimensions of pain interference have varying effects on the dimensions of quality of life. These insights advocate for a multidimensional approach to pain management in cancer care to improve patient quality of life.

Keywords Cancer pain, Quality of life, Biopsychosocial model, Structural equation model

*Correspondence:

Yongchun Cui
yccui@sdfmu.edu.cn

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Introduction

Cancer pain represents a major global public health challenge. Although this study employs the concise clinical term “cancer pain” for practical purposes, the methodology strictly conforms to the International Association for the Study of Pain (IASP) diagnostic criteria for chronic cancer-related pain as defined in ICD-11. Specifically, chronic cancer-related pain is defined as chronic pain caused by the primary cancer itself or its metastases (referred to as chronic cancer pain), as well as chronic pain resulting from cancer treatment (known as chronic postcancer treatment pain) [1]. A meta-analysis revealed that the overall prevalence of pain among cancer patients was 44.5%, with 30.6% experiencing moderate to severe pain [2]. Pain persists throughout the continuum of cancer care, imposing a sustained therapeutic burden on patients. A longitudinal study tracking cancer patients from diagnosis through one-year follow-up revealed significant temporal reductions in depression and anxiety levels, whereas pain severity demonstrated no statistically significant improvement [3].

In 1995, the World Health Organization (WHO) established quality of life as a multidimensional construct encompassing individuals’ subjective evaluation of their physical, psychological, and social functioning within cultural and environmental contexts, specifically addressing personal goal attainment, value systems, and existential concerns. A meta-analysis found that quality of life is significantly impacted 2 to 26 years after cancer diagnosis, especially in physical and mental health [4]. Contemporary oncology has witnessed transformative therapeutic advances through evidence-based precision medicine, enabling unprecedented extension of overall survival durations in cancer populations [5, 6]. The paradigm of oncological care has progressively evolved from a singular emphasis on survival prolongation to a more holistic approach prioritizing quality of life optimization [4, 7, 8]. However, this therapeutic success introduces new challenges, as prolonged survival often correlates with sustained treatment-related morbidity and pain syndromes. The clinical assessment of pain presence and severity serves as a prognostic indicator for quality of life [9]. Evidence demonstrates that enhanced pain symptom monitoring improves life quality, reduces healthcare utilization, and enhances adherence to antitumor therapies [10].

Pain is a critical factor contributing to diminished quality of life across diverse patient populations, with cancer pain associated with significantly poorer quality of life compared to other disease groups [11]. Physical, psychological, and social pain—three interrelated and mutually influential dimensions—collectively shape an individual’s well-being. Physical pain, a defensive response to noxious stimuli, activates nociceptive pathways in the peripheral

and central nervous systems, leading to somatic discomfort. Chronic pain further disrupts sleep, limits daily activities, and compromises immune function, with cancer patients experiencing pain reporting markedly lower overall quality of life than their pain-free counterparts [12]. Psychological pain manifests as emotional disturbances, including depression, affective distress, fear, anxiety, irritability, impaired concentration, and persistent worry. A multinational study of cancer patients in Canada and Europe revealed that over 70% of advanced-stage patients endure intolerable pain, which acts as a severe stressor and negative life event, triggering psychological disorders in approximately 50% of cases and substantially degrading quality of life [13]. Social pain, defined as the negative emotional experience arising from threats to or disruptions of social bonds or self-worth, diminishes individuals’ sense of belonging, control, and meaningful existence, fostering learned helplessness and social withdrawal that further impair quality of life [14]. While extant research on the pain-quality of life relationship in cancer patients provides substantive clinical guidance, two critical knowledge gaps persist in contemporary scholarship: ①The predominant operationalization of pain and quality of life as composite measures, neglecting multidimensional decomposition analyses. ②Methodological constraints yielding compromised generalizability of findings. The study aims to establish a detailed framework for the relationship between pain and quality of life.

The biopsychosocial model provides a robust analytical framework for investigating pain and quality of life in clinical research. Originating from George Engel’s seminal 1977 work introducing the “biopsychosocial approach,” this paradigm shift transcends biomedical reductionism by conceptualizing health outcomes as emergent properties of dynamic interactions between biological, psychological, and social determinants [15]. Within this framework, Floriana et al. [7] proposed the concept of quality of life from a biopsychosocial perspective, and Gatchel et al. [16] introduced the Biopsychosocial Model of Chronic Pain to explain and understand the causes and impacts of chronic pain. Pain and quality of life were assessed using two widely validated scales in cancer populations: the Brief Pain Inventory-Short Form (BPI-SF) and the Quality of Life Questionnaire-Function 17 (QLQ-F17). Both instruments align with the biopsychosocial framework, capturing multidimensional interactions. The BPI-SF quantifies not only subjective pain intensity but also its interference with daily activities (e.g., mobility, work) and psychosocial well-being (e.g., mood, relationships, sleep). Similarly, the QLQ-F17 evaluates physical dysfunction alongside role, cognitive, emotional, and social functioning. This study aims to measure the multidimensional impact of cancer pain on quality of

life. Our two prespecified hypotheses are: H1—Each pain dimension (intensity, affective interference, activity interference) independently reduces physical, role, emotional, cognitive, and social functioning; H2—Comprehensive pain perception negatively impacts these five functions of quality of life.

SEM integrates latent constructs with observed variables, accounts for measurement errors, and simultaneously tests multivariate relationships through unified equation systems. Notably, it synthesizes traditional methodologies (e.g., regression, path, and factor analyses) while providing robust fit indices for hypothesis validation [17, 18]. Critically, SEM enables systematic quantification of bidirectional relationships among biological, psychological, and social pain dimensions, mapping their distinct impacts on physical, emotional, and social domains of quality of life. This approach aligns with its recognition as a paradigm-shifting advancement in applied statistics, offering unparalleled precision in modeling complex health outcomes.

Materials and methods

Participants

This cross-sectional study recruited participants experiencing chronic cancer-related pain using convenience sampling from a tertiary cancer hospital in Shandong Province, China, between January and February 2024. Eligible patients meeting the predefined inclusion and exclusion criteria were surveyed through face-to-face questionnaires. Ethical approval for this study was obtained from the Ethics Committee of Shandong Cancer Hospital (Approval No. SDTHEC2024004030).

Inclusion and exclusion criteria

Inclusion criteria: ① Age > 18 (years) ② Signed informed consent; ③ Histologically confirmed cancer diagnosis; ④ Chronic cancer-related pain (classified under ICD 11: MG30.5 based on etiology) [19]; ⑤ Absence of severe psychiatric or physical disorders affecting communication.

Exclusion criteria: ① Non-cancer patients or patients without pain; ② Pain unrelated to cancer; ③ Incomplete, logically inconsistent, or contradictory questionnaire data; ④ Comorbidities severely interfering with cancer pain assessment (e.g., fractures, neuropathic pain); ⑤ Refusal to cooperate or inability to communicate.

Instruments

General participant characteristics

Demographic and clinical data were obtained through a combination of self-administered questionnaires and electronic medical records. The collected variables included: ① demographic characteristics (gender, age); ② disease profiles (cancer type, disease stage, metastasis, pain location, comorbidities); and ③ treatment history

(surgery, radiotherapy, chemotherapy, targeted therapy, immunotherapy).

BPI-SF

The BPI-SF, a widely recognized multidimensional pain assessment tool, was used to evaluate the pain experienced by the cancer patients over the past 24 h. This instrument, which has been adapted into Chinese and validated for its internal consistency and test-retest reliability [20–22]. It exhibits two commonly recognized factor structures: ① Two-factor structure: Comprising “pain intensity” and “pain interference”; ② Three-factor structure: Encompassing “pain intensity,” “activity interference,” and “affectivity interference.” It utilizes an 11-point rating scale ranging from 0 (“no pain”) to 10 (“pain as bad as you can imagine”).

QLQ-F17

The QLQ-F17, developed by the European Organization for Research and Treatment of Cancer (EORTC), is designed to assess patients’ quality of life over the past week and includes only the functional scales and the self-rated health scale from the QLQ-C30 [23]. Translated into multiple languages and widely used globally, this tool has demonstrated strong internal consistency in the Chinese cancer population (Cronbach’s $\alpha > 0.70$) [24]. It consists of five functional scales: Physical Function, Role Function, Cognitive Function, Emotional Function, and Social Function, along with one item for patient self-rated health. Functional scales are rated on a four-point scale (1 = “No functional impairment” to 4 = “Severely impaired”). Self-rated health is measured on a seven-point scale (1 = “Poor” to 7 = “Excellent”), with higher scores indicating better health. Scores are transformed linearly to a 0–100 scale, with higher values indicating better function and health.

Sample size calculation

The sample size for SEM is generally recommended to exceed 200 or be 10 times the number of parameters [17, 18]. In this study, variables included twelve General Participant Characteristics items, three domains from the BPI-SF, and six domains from the QLQ-F17, totaling 21 variables for statistical analysis. The minimum required sample size was calculated as 10 times the number of parameters ($21 \times 10 = 210$). To ensure robustness, 300 questionnaires were distributed, and 276 valid responses were collected, meeting the minimum sample size requirement.

Quality control

To ensure methodological rigor, a four-phase quality control protocol was implemented: ① Pre-survey standardization: Investigators underwent systematic training to

align understanding of research objectives, data collection procedures, and operational definitions; ② Participant screening: Strict adherence to predefined inclusion/exclusion criteria was maintained, with written informed consent obtained after full disclosure of study purposes; ③ Post-survey verification: All collected data underwent dual-entry validation and completeness checks; questionnaires with missing responses, logical inconsistencies, or protocol deviations were excluded; ④ Analytical robustness: During analysis of SEM, demographic characteristics, disease profiles and treatment history were incorporated as control variables to mitigate confounding effects.

Analytical methods

Descriptive statistics and reliability tests were performed using SPSS 27.0. Specifically, descriptive statistics were conducted for demographic characteristics, disease - related information, BPI - SF scores, and QLQ - F17 scores. Categorical variables were represented by composition ratios, and continuous variables were shown

as $\bar{x} \pm s$. The scale reliability was assessed using Cronbach's α (>0.60 accept, >0.70 good, >0.80 better, >0.90 best) [25]. Then, confirmatory factor analysis (CFA) and SEM construction were carried out using AMOS 23.0. As a sub - model of SEM, CFA is a special application for SEM analysis. In this study, CFA was used to evaluate the structural validity of each dimension. Combinatorial reliability (CR) and average variance extracted (AVE) were calculated for this purpose. According to the criteria [17, 18, 26, 27], $AVE > 0.36$ is acceptable, and $AVE \geq 0.50$ is considered excellent. Standardized factor loadings were estimated to assess the relationships between latent and observed variables by SEM, as well as the interactions among latent variables. The model fit was evaluated using several fit indices: Chi-square/df ratio (χ^2/df): ideally between 1 and 3. Goodness-of-Fit Index (GFI) and Adjustment of Goodness-of-Fit Index (AGFI): >0.80 is acceptable, and >0.90 is preferable. Root Mean Square Error of Approximation (RMSEA): ≤ 0.08 . Standardized Root Mean Square Residual (SRMR): <0.50 . Normative Fit Index (NFI), Tucker-Lewis Coefficient (TLI)/Non-normed Fit Index (NNFI), and Comparative Fit Index (CFI): >0.90 . If the model meets the criteria [17, 18], it is considered to have an acceptable fit with the sample data. If it fails to meet these criteria, theoretical model revisions are needed.

The SEM consists of two parts: the measurement model and the structural model. The SEM parameter estimation was performed using the Maximum Likelihood Estimation (MLE) method [18] with control for general demographic and disease - related variables to account for confounding.

Measurement model

The measurement model was defined by two linear equations. Equation ① describes the relationship between exogenous latent variables and their corresponding observed variables, while Equation ② specifies the relationship between endogenous latent variables and their observed variables. The equations are expressed as follows:

$$X = \Lambda_x \xi + \sigma \quad (1)$$

$$Y = \Lambda_y \eta + \epsilon \quad (2)$$

Here, X and Y represent the exogenous and endogenous observed variables, respectively; Λ_x and Λ_y denote the factor loading matrices that map the relationships between the observed variables and their corresponding latent variables ξ for exogenous and η for endogenous; σ and ϵ are the measurement errors associated with the exogenous and endogenous observed variables.

Structural model

The structural model characterizes the interrelationships among latent variables and is formulated as:

$$\eta = B_\eta + T\xi + \zeta \quad (3)$$

In this equation, B represents the matrix of coefficients describing the relationships between endogenous latent variables, T quantifies the effects of exogenous latent variables on endogenous latent variables, and ζ denotes the error term of the structural equation.

Results

Participant characteristics

In this study, 300 questionnaires were distributed, and 276 valid responses were collected, resulting in a valid response rate of 92.00%.

The participant demographics are summarized in Table 1. The study predominantly included male patients, accounting for 57.97%, with females making up 42.03% of the sample. The majority of participants were in the age range of 40 to 60 years, at 52.90%. The most frequently reported types of cancer were respiratory and digestive system cancers, accounting for 36.96% and 38.04% of cases, respectively. Stage IV cancer was the most common, representing 39.49% of the cohort. In terms of pain severity, 69.57% of participants reported mild pain, 22.10% moderate pain, and 8.33% severe pain. The most common locations of pain reported were abdominal (36.23%), chest (28.62%), and back (21.01%). Additionally, 46.70% of patients had metastasis. Among the treatments received, 65.22% had undergone surgery, 53.62% chemotherapy, 28.62% targeted therapy, and 21.38%

Table 1 Basic characteristics of study participants (n = 276)

Characteristics	Category	Number	Percent(%)
Gender	Male	160	57.97
	Female	116	42.03
Age (years)	<40	18	6.52
	40–60	146	52.90
	>60	112	40.58
Cancer type	Respiratory system cancers Tumors	102	36.96
	Digestive system cancers	105	38.04
	Head and neck cancers	20	7.25
	Urogenital system cancers	32	11.59
	others	17	6.16
Cancer stage	I	56	20.29
	II	37	13.41
	III	56	20.29
	IV	109	39.49
	Unknown	18	6.52
metastasis	Yes	16	5.80
	No	129	46.70
	Unknown	131	47.50
Pain intensity	Mild	192	69.57
	Moderate	61	22.13
	Severe	23	8.33
Pain locations	Head and neck	46	16.67
	Scapular region	23	8.33
	Chest	79	28.62
	Abdomen	100	36.23
	Back	58	21.01
	Hip	31	11.23
	Upper limb	5	1.81
	Lower limb	24	8.70
	Unknown	16	5.80
Surgery	Yes	180	65.22
	No	96	34.78
Chemotherapy	Yes	148	53.62
	No	128	46.38
Radiotherapy	Yes	86	31.16
	No	190	68.84
Targeted Therapy	Yes	79	28.62
	No	197	71.38
Immunotherapy	Yes	59	21.38
	No	217	78.62
Comorbidities	Yes	108	39.13
	No	168	60.87

immunotherapy. Furthermore, 39.13% of patients had comorbidities or underlying diseases, while 95.30% had no family history of cancer.

Scores of brief pain Inventory-short form

The study subjects scored 3.62 ± 1.78 on the pain intensity dimension, 3.72 ± 2.57 on the pain interference with

Table 2 Scores of brief pain inventory-short form

Variables	Min	Max	$\bar{x} \pm s$	Skewness	Kurtosis
Pain	1	10	3.90 ± 2.13	0.593	-0.423
Pain intensity	1	10	3.62 ± 1.78	0.736	0.358
Worst Pain	1	10	6.58 ± 2.59	-0.099	-1.265
Least Pain	1	10	2.09 ± 2.08	1.311	2.160
Average Pain	1	10	2.96 ± 1.89	1.122	0.834
Pain Now	1	10	2.85 ± 2.31	0.811	0.321
activity interfere	1	10	3.72 ± 2.57	0.564	-0.609
General Activity	1	10	4.92 ± 3.16	0.389	-1.113
Walking	1	10	4.26 ± 3.52	0.549	-1.156
Working	1	10	4.36 ± 3.48	0.461	-1.165
affectivity interfere	1	10	4.51 ± 3.11	0.502	-1.107
Mood	1	10	3.91 ± 3.07	0.584	-0.699
Relations with Others	1	10	2.59 ± 2.59	1.138	0.691
Sleep	1	10	4.94 ± 3.34	0.119	-1.241
Enjoyment of Life	1	10	3.42 ± 3.00	0.799	-0.265

Note: Mild pain [1–4], Moderate pain ([5]– [6]), Severe pain [7–10]. Descriptive statistics for pain intensity are presented in Table 1

Table 3 Scores of quality of life Questionnaire-Function 17

Variables	Min	Max	$\bar{x} \pm s$	Skewness	Kurtosis
Quality of Life	0	100	54.48 ± 19.16	-0.315	-0.594
Physical Function	0	100	55.10 ± 33.19	-0.305	-1.268
Role Function	0	100	41.43 ± 31.48	-0.039	-1.221
Emotional Function	0	100	43.18 ± 29.59	-0.964	0.839
Cognitive Function	0	100	77.20 ± 21.71	-0.323	-1.311
Social Function	0	100	52.36 ± 31.16	-0.237	-0.837
self-rated health	0	100	57.61 ± 23.18	-0.072	-0.648

activity dimension, and 4.51 ± 3.11 on the pain interference with emotion dimension. See Table 2 for more details.

Scores of quality of life questionnaire-function 17

The study subjects scored 77.20 ± 21.71 on the cognitive function, followed by self-rated health at 57.61 ± 23.18 . Role functioning and emotional functioning were lower, at 41.43 ± 31.48 and 43.18 ± 29.59 respectively. See Table 3 for more details.

Brief pain inventory models

As shown in Table 4, the overall pain scale demonstrated excellent internal consistency with a Cronbach's α of 0.918. The pain intensity subscale Cronbach's α was 0.808. For the pain interference dimension, the Cronbach's α was 0.922. Within this dimension, the pain interference affectivity subscale had a Cronbach's α of 0.905, and the pain interference activity subscale had a Cronbach's α of 0.874.

The definitions of M1–M5 are detailed in Fig. 1. As shown in Table 5, CR values >0.7 and AVE values >0.50 for M1–M5. Notably, M2 exhibited superior psychometric properties compared to other models: pain intensity

Table 4 Brief pain inventory short form scale reliability evaluation

Variables	Cronbach's α
Pain (BPI3-BPI6, BPI9a-BPI9g)	0.918
pain intensity (BPI3-BPI6)	0.808
Pain interference (BPI9a-BPI9g)	0.922
Pain interference activity (BPI9a、9c、9d)	0.874
Pain interference affectivity (BPI9b、9e、9f、9 g)	0.905

Note: Criteria [25]: Cronbach's α >0.60 accept, > 0.70 good, > 0.80 better, > 0.90 best

(CR=0.886, AVE=0.660), pain interference with activity (CR=0.906, AVE=0.763), and pain interference with

affectivity (CR=0.879, AVE=0.710). M3 and M5 displayed identical CR and AVE values, indicating equivalent measurement precision.

As shown in Table 6, CFA revealed that RMSEA and SRMR values for M1–M5 fell outside recommended thresholds. Meanwhile, among the remaining fit indices, M2, M3, and M5 showed better alignment with the Model fit criteria compared to M1 and M4. M2, M3, and M5 shared identical CFA model fit indices, as did M1 and M4, respectively, indicating equivalent measurement precision.

M2 demonstrated superior fit. M3 and M5 showed similar CFA model fit indices, but M3 was more

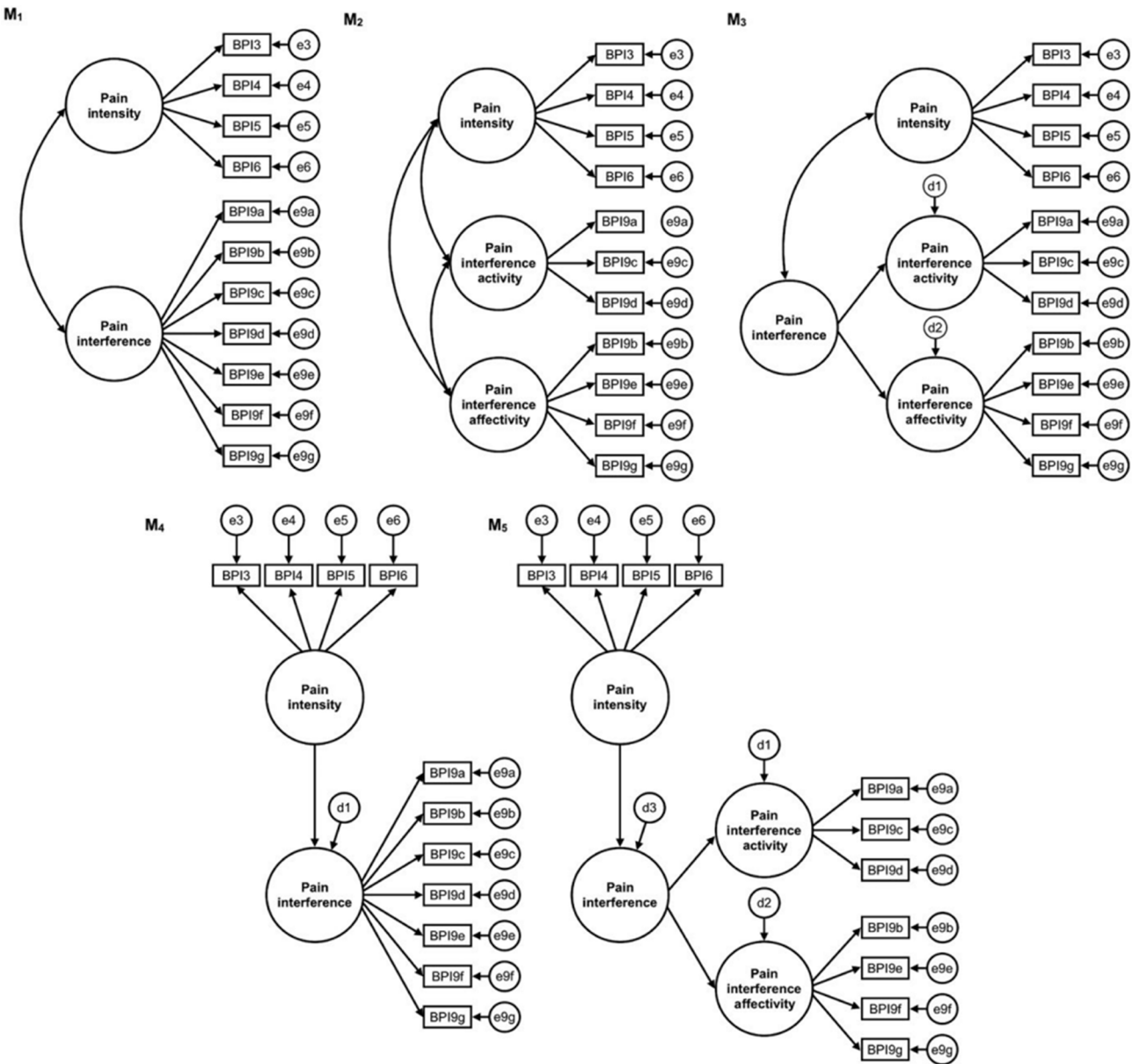


Fig. 1 Theoretical models of the Brief Pain Inventory

Table 5 Average variance extracted and reliability of the brief pain inventory models

Pain category	M1		M2		M3		M4		M5	
	CR	AVE	CR	AVE	CR	AVE	CR	AVE	CR	AVE
intensity	0.793	0.490	0.886	0.660	0.821	0.536	0.821	0.536	0.821	0.536
interference	0.924	0.637					0.924	0.637		
activity interfere			0.906	0.763	0.906	0.763			0.906	0.763
affectivity interfere			0.879	0.710	0.879	0.645			0.879	0.645

Note: Abbreviations are defined in the list at the end of the article. Criteria: CR ≥ 0.70 fine; AVE > 0.36 accept, ≥ 0.50 fine

Table 6 Confirmatory factor analysis of the brief pain inventory models

Model fit indices	Criteria (17, 18)	M1	M2	M3	M4	M5
χ^2/df	1 ~ 3	5.666	4.135	4.135	5.666	4.135
GFI	> 0.80 accept, > 0.90 fine	0.857	0.897	0.897	0.857	0.897
AGFI	> 0.80 accept, > 0.90 fine	0.780	0.834	0.834	0.780	0.834
RMSEA	≤ 0.08	0.130	0.107	0.107	0.130	0.107
SRMR	< 0.50	0.057	0.052	0.052	0.057	0.052
NFI	> 0.90	0.879	0.915	0.915	0.879	0.915
TLI (NNFI)	> 0.90	0.868	0.912	0.912	0.868	0.912
CFI	> 0.90	0.897	0.934	0.934	0.897	0.934

Note: Abbreviations are defined in the list at the end of the article

Table 7 Model fit indices in structural equation model

Model fit indices	criteria(17, 18)	One-phase	Two-phase	Adjusted two-phase
χ^2/df	1 ~ 3	2.653	2.950	2.691
GFI	> 0.80 accept, > 0.90 fine	0.826	0.805	0.826
RMSEA	< 0.08 fine, $0.08 \sim 0.10$ accept	0.078	0.084	0.078
TLI (NNFI)	> 0.90	0.902	0.885	0.900
CFI	> 0.90	0.916	0.897	0.912

Note: Abbreviations are defined in the list at the end of the article

parsimonious, with bidirectional associations between pain intensity and pain interference eliminating residual estimation needs. Moreover, M3 is more widely used in both research and clinical settings, enhancing result interpretation. Overall, we selected M2 and M3 to construct a SEM of pain and quality of life. For M2 and M3, Model fit indices were as follows: GFI=0.897, AGFI=0.834, NFI=0.915, TLI=0.912, and CFI=0.934.

Source of image: brief pain inventory: a proposal to extend its clinical application [26]

Note: pain intensity (items BPI3-BPI6); pain interference (items BPI9a-BPI9g); Pain interference activity (items BPI9a, 9c, 9d); Pain interference affectivity (items BPI9b, 9e, 9f, 9g).

M1: Bifactorial model: 11 items distributed into two factors: “Pain intensity” and “Pain interference with daily functioning”.

M2: Trifactorial Model: 11 items distributed into three factors “Pain intensity”, “Interference of pain in daily activities” and “Interference of pain in affectivity”.

M3: Trifactorial Model with second-order hierarchical factor: 11 items distributed into three factors “Pain intensity”, “Interference of pain in activities”, “Interference of

pain in affectivity” and second-order hierarchical factor called “Pain interference,” which reflects the two latter factors.

M4: Structural Model proposed from the hypothetical causal relationship of “Pain intensity” in the “Pain interference with daily functioning,” considering the “Pain interference with daily functioning” a first-order factor (as per M1).

M5: Structural Model proposed from the hypothetical causal relationship of “Pain intensity” in “Pain interference,” considering “Pain interference” as a second-order hierarchical factor (as per M3).

One-Phase SEM between pain and quality of life

As shown in Table 7, the fit indices of the One-Phase SEM included a χ^2/df of 2.653, GFI of 0.826, RMSEA of 0.078, TLI of 0.902, and CFI of 0.916, indicating a satisfactory model fit according to established criteria. As shown in Table 8; Fig. 2, cognitive function was significantly affected by pain intensity ($B=0.165$, $P<0.001$). Other dimensions of quality of life did not show significant changes with pain intensity alone. Meanwhile, pain interference with activity significantly influenced physical function ($B=0.576$), role function ($B=0.585$), cognitive

Table 8 Estimated coefficients of one phase structural equation model weights

Relationships of latent variables	Estimate (B)	SE	CR	P	Standardized estimates
Pain intensity→PF	0.053	0.055	0.954	0.340	0.101
Pain intensity→RF	0.047	0.056	0.846	0.398	0.087
Pain intensity→EF	0.015	0.043	0.340	0.734	0.038
Pain intensity → CF	0.165	0.039	4.290	<0.001	0.327
Pain intensity → SF	-0.012	0.055	-0.214	0.831	-0.021
Pain interference activity→PF	0.576	0.068	8.433	<0.001	1.830
Pain interference activity→RF	0.585	0.065	9.030	<0.001	1.793
Pain interference activity→EF	0.068	0.045	1.514	0.130	0.290
Pain interference activity→CF	0.273	0.039	7.010	<0.001	0.900
Pain interference activity→SF	0.260	0.058	4.482	<0.001	0.774
Pain interference affectivity→PF	-0.467	0.080	-5.838	<0.001	-1.351
Pain interference affectivity→RF	-0.455	0.078	-5.867	<0.001	-1.272
Pain interference affectivity→EF	0.002	0.055	0.033	0.974	0.007
Pain interference affectivity→CF	-0.084	0.046	-1.816	0.069	-0.251
Pain interference affectivity→SF	-0.129	0.070	-1.843	0.065	-0.350

Note: Abbreviations are defined in the list at the end of the article

function ($B=0.273$), and social function ($B=0.260$) (all $P<0.001$). Pain interference with affectivity significantly improved physical function ($B = -0.467$) and role function ($B = -0.455$) (both $P<0.001$), but not other quality of life dimensions.

Two-Phase SEM between pain and quality of life

As shown in Table 7, the fit indices of the Two-Phase SEM initially showing a χ^2/df of 2.950, GFI of 0.805, RMSEA of 0.084, TLI of 0.885, and CFI of 0.897, required adjustments for better fit based on modification indices. Based on modification indices (MI), residual correlations with significant impacts on model fit were adjusted. The MI for residuals e43–e44 was 28.129 (parameter change value = 1.182), for e40–e47 it was 25.051 (parameter change value = 1.465), and for e20–e21 it was 14.047 (parameter change value = 0.105). After freeing the covariances between these residuals (e43–e44, e40–e47, and e20–e21), the model fit improved substantially: the RMSEA decreased to 0.069, the TLI increased to 0.900, and the CFI rose to 0.912. As shown in Table 9; Fig. 3, in this refined model, pain as a two-phase latent variable significantly impacted physical function ($B=0.480$), role function ($B=0.540$), emotional function ($B=0.195$),

cognitive function ($B=0.715$), and social function ($B=0.369$), all significant at $P<0.001$.

Discussion

This study employed a biopsychosocial framework to develop SEMs quantifying the multidimensional impact of pain on quality of life in cancer patients, aiming to provide scientific evidence and novel clinical perspectives for evidence-based pain management strategies. From the five theoretical models of the BPI, the two most representative pain models were selected. These were integrated with the five functional domains derived from the QLQ-F17 to construct two SEMs: a one-phase model and a two-phase model. H1 was supported except for the relationships between pain interference affectivity and physical function, as well as role function, as determined through one-phase SEM. H2 was fully supported through two-phase SEM. The results of the one-phase SEM showed that distinct pathways through which pain dimensions impact quality of life. The results of the two-phase SEM indicated that pain affects all aspects of the quality of life functional domains. These results provide important clues for further exploring the mechanisms by which pain affects quality of life.

Consistent with Bonafé et al. [28], our analyses confirmed the superiority of the BPI's three-factor structure (pain intensity, activity interference, and affectivity interference) over the traditional two-factor model in terms of psychometric validity and clinical interpretability. However, when modeling the integrated relationships between pain dimensions and quality-of-life domains, the second-order structural equation model provided a more comprehensive theoretical framework. This hierarchical configuration demonstrated greater explanatory power for interpreting clinical findings, suggesting its utility in guiding patient-centered pain management and quality-of-life enhancement strategies. Notably, while the first-order model exhibited marginally superior fit indices statistically, its interpretation requires caution—particularly regarding the complex relationships between pain interference affectivity and physical/role functioning. Further validation of these specific pathways remains warranted.

Consistent with findings from John et al. [29] and Karayannis et al. [30], our results confirm that pain has a negative impact on all aspects of the quality of life functional domains, including physical function, role function, emotional function, cognitive function, and social function. Moreover, multiple studies highlight the potential of cancer pain to affect all dimensions of quality of life [31, 32]. As pain intensity increases, individuals report negative impacts on daily work and activities, emotions, spirits, sleep, and diet, with gender differences noted [33]. Specifically, cancer patients often report inability

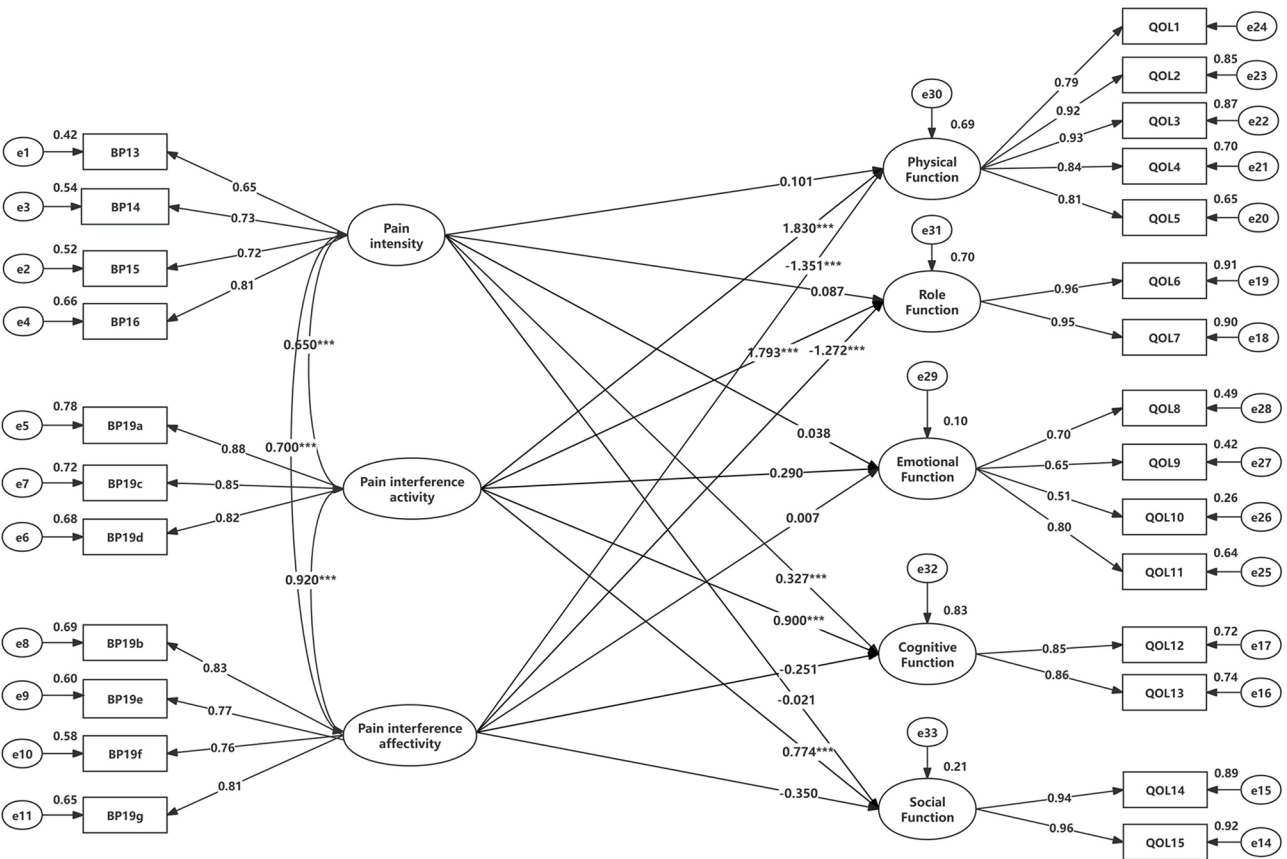


Fig. 2 One phase structural equation model of pain and quality of life Note: The correlation coefficients among latent variables,*** $P<0.001$. In SEM, observed variables refer to directly measurable or observable data, typically represented as individual measurement items in questionnaires and graphically depicted as squares. Latent variables, on the other hand, represent the underlying factors (dimensions) extracted from the scales and are conventionally illustrated as ellipses in the model. We constructed a SEM incorporating three pain factors from the BPI-SF (pain model 2 as shown in Fig. 1) and five functional domains from the QLQ-F17 as latent variables, with their respective scale items serving as observed variables

Table 9 Estimated coefficients of two-phase structural equation model weights

Relationships of latent variables	Estimate (B)	SE	CR.	P	Standardized estimates
pain→intensity	1.000				0.688
pain → activity interfere	2.560	0.295	8.677	<0.001	0.992
pain → affectivity interfere	1.585	0.202	7.860	<0.001	0.871
pain → PF	0.480	0.070	6.868	<0.001	0.617
pain → RF	0.540	0.073	7.379	<0.001	0.658
pain → EF	0.195	0.046	4.205	<0.001	0.327
pain → CF	0.715	0.086	8.352	<0.001	0.921
pain → SF	0.369	0.065	5.720	<0.001	0.441

Note: Abbreviations are defined in the list at the end of the article

to walk or stand due to pain [34–36], severely affecting their daily living and self-care abilities. The dual negative effects of cancer and pain disrupt individual occupational and familial roles [11], reducing social activities [31]. Moreover, cancer pain significantly affects attention, memory, and executive functions in patients [37, 38], also

influencing emotional states [39] which studies indicate higher levels of anxiety and depression among cancer patients with pain compared to pain-free cancer patients. The positive impact of pain interference affectivity on physical and role function in this study can be explained in two ways: Firstly, this counterintuitive result may be due to factors such as catastrophic thinking, anxiety, and depression. These factors affect emotional and cognitive states, but do not significantly impact physical capacity or role performance [40, 41]. Secondly, the BPI-SF and QLQ-F17 measure different time spans. Pain interference with emotions measured by the BPI-SF is based on pain experiences in the past 24 h, while the QLQ-F17 assesses functional states based on life perceptions in the past week. This means that although individuals may have experienced pain interference with emotions in the past 24 h, their physical and role function may have improved within the past week. The QLQ-F17 self-rated health score of 57.61 ± 23.18 ranks second in the quality of life dimensions and shows a relatively good level. Looking at the raw data, 200 of the 276 patients had a

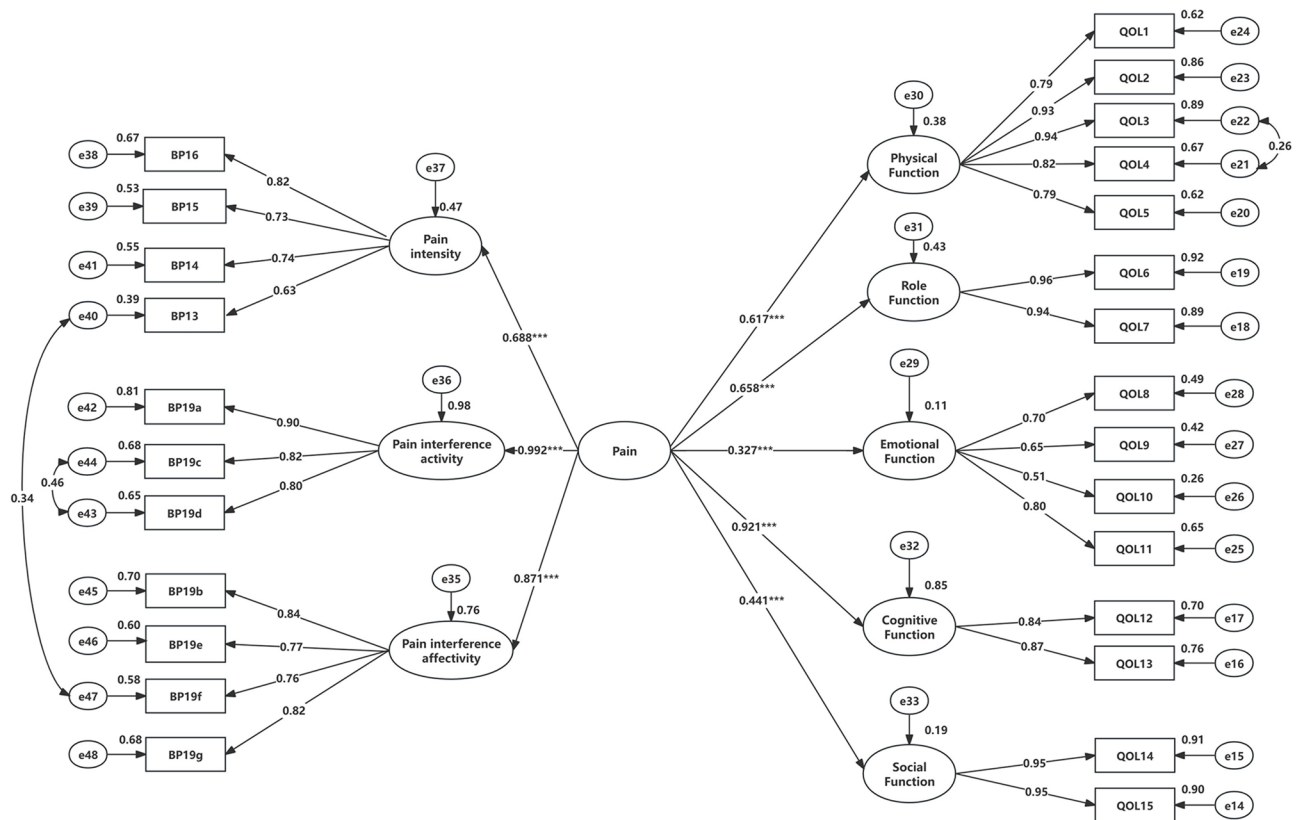


Fig. 3 Two-phase structural equation model of pain and quality of life Note: The correlation coefficients among latent variables, *** $P < 0.001$. In SEM, observed variables refer to directly measurable or observable data, typically represented as individual measurement items in questionnaires and graphically depicted as squares. Latent variables, on the other hand, represent the underlying factors (dimensions) extracted from the scales and are conventionally illustrated as ellipses in the model. We constructed a SEM incorporating pain as a two-phase latent variable (pain model 3 as shown in Fig. 1) and five functional domains from the QLQ-F17 as latent variables, with their respective scale items serving as observed variables

quality of life score of ≥ 50 (out of 100), which indirectly confirms that patients perceived their physical condition to have improved compared to a week ago. The reasons for this improvement are multi-factorial, including physiological, psychological, and social aspects. Physiologically, comprehensive treatment can restore health. Psychologically, positive coping with the disease and post-traumatic growth can enhance psychological resilience [42], thereby improving patients' quality of life and functional status. Socially, increased self-support [43] and social support [44] can buffer the impact of negative events. Therefore, even though pain has a negative effect on emotions, it can positively promote physical and role function under the overall effect.

Human neuroimaging research has consistently sought to characterize how cognitive factors modulate neural processing and pain perception. According to the top-down pain regulation framework [45], nociceptive input triggers a perceptual system that compares actual sensory signals with prior expectations. When pain perception aligns with expectations, those expectations are reinforced; discrepancies generate prediction error signals that update future expectations through learning

mechanisms. The pain matrix comprises two functionally distinct subsystems: a sensory-discriminative component and a cognitive-affective system. The latter, involving regions like the anterior insula and anterior cingulate cortex, primarily mediates psychological dimensions of pain [45]. Crucially, cognitive factors—including attentional allocation, expectancy, catastrophizing, reappraisal, and perceived control—dynamically regulate pain perception. Modifying maladaptive cognitive patterns (e.g., reducing pain catastrophizing) can ameliorate suffering, as exemplified by placebo analgesia and attentional disengagement phenomena [46]. However, pathological conditions may impair predictive error processing, leading to aberrant learning and persistent pain states. This is evidenced by increased error rates and prolonged reaction times during Stroop tasks among pain patients, particularly during acute episodes [45]. Consequently, cognitive and psychological interventions offer promising avenues for restoring adaptive pain regulation.

This study demonstrates that effective cancer pain management requires a multidimensional biopsychosocial approach, implementing integrated interventions to holistically improve all functional domains of patients'

quality of life. Based on our findings, we propose the following evidence-based strategies: First, comprehensive multidimensional pain management should address not only physiological sensations but also emotional, cognitive, and social dimensions. Beyond conventional pharmacotherapy, interventions must incorporate cognitive-behavioral therapy (CBT) [47] and robust social support systems [48, 49]. For cognitive-emotional regulation, healthcare teams should provide structured training in pain coping strategies—including relaxation techniques, CBT, and adaptive skills—to modify maladaptive pain cognitions and enhance emotional regulation, thereby improving life satisfaction. Concurrently, building social support networks necessitates addressing patients' emotional needs through psychological counseling, rehabilitation teams, and support groups that provide emotional validation and informational guidance. Facilitating open communication with family and friends strengthens these networks while empowering patients through self-monitoring and pain management skill development. Second, personalized integrated treatment plans should target improvements across all five quality-of-life functional domains through coordinated pharmacological, psychological, social, and physical therapies. This requires establishing longitudinal partnerships between healthcare teams and patients, utilizing Two-Phase SEM methodology to periodically assess pain trajectories and quality of life metrics. By maintaining comprehensive health records and enabling dynamic treatment adjustments, this approach ensures therapeutic adaptability and maximizes overall health-related quality of life outcomes.

Limitations and future directions

This study employed SEM to comprehensively assess the multidimensional impact of cancer-related pain, offering methodological rigor and novel insights. Nevertheless, several limitations warrant acknowledgment. First, the cross-sectional design captures variable relationships at a single time point, precluding causal inferences and longitudinal dynamics. Second, the single-center convenience sampling approach may introduce selection bias, potentially limiting the generalizability of findings. Third, the complexity of cancer pain etiology was not fully addressed, as pain management strategies (e.g., pharmacological/non-pharmacological interventions) were not systematically controlled, potentially confounding observed relationships. Critically, while the SEM demonstrated excellent model fit, it lacks external validation using independent cohorts, thereby constraining the extrapolation of results.

Conclusion

This research elucidates that pain and quality of life are intrinsically linked through complex physiological, psychological, and social interactions. Pain adversely affects multiple domains of life, necessitating a comprehensive clinical approach to pain management that addresses its multifaceted nature to enhance overall quality of life. Our findings advocate for the integration of pain monitoring and tailored patient-centered strategies in clinical settings.

Abbreviations

BPI-SF	Brief Pain Inventory Short Form
QLQ-F17	Quality of Life Questionnaire-Function 17
QOL	Quality of Life
PF	Physical Function
RF	Role Function
EF	Emotional Function
CF	Cognitive Function
SF	Social Function
CFA	Confirmatory Factor Analysis
MLE	Maximum Likelihood Estimation
SEM	Structural Equation Modeling
χ^2/df	Chi-square/df ratio
GFI	Goodness-of-Fit Index
AGFI	Adjustment of Goodness-of-Fit Index
RMSEA	Root Mean Square Error of Approximation
SRMR	Standardized Root Mean Square Residual
NFI	Normative Fit Index
TLI/NNFI	Tucker-Lewis Coefficient/Non-normed Fit Index
CFI	Comparative Fit Index
CR	Combinatorial Reliability
AVE	Average Variance Extracted
SE	Standard Error
CR	Critical Ratio

Acknowledgements

We acknowledge funding by Shandong Province Humanities and Social Sciences Special Fund for Health Education and Health Promotion (2023-JKZX-08); Shandong Province Medical and Health Science and Technology Project (202315020614).

Author contributions

Xiaoyu Hu conceptualised the study and completed the first draft of the manuscript. Yongchun Cui was responsible for the decision to submit the manuscript. Jingjing Lv, Chang Qin, Xiaoxuan Liu, Jianbiao Zhu, Yongchun Cui provided critical comments on drafts of the manuscript. All authors participated in the review of the manuscript and read and approved the final manuscript. All authors had access to the data in the study and had final responsibility to submit for publication.

Funding

Shandong Province Humanities and Social Sciences Special Fund for Health Education and Health Promotion (2023-JKZX-08); Shandong Province Medical and Health Science and Technology Project (202315020614).

Data availability

The data used for the analyses are available by email request to the corresponding author.

Declarations

Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki and was approved by the Ethics Committee of Shandong Cancer Hospital (approval number: SDTHEC2024004030). Written informed consent was obtained from all participants prior to their inclusion in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Shandong Cancer Hospital and Institute, Shandong First Medical University, Shandong Academy of Medical Sciences, Jinan, Shandong, China

²Expanded Program Immunization Division of Shandong Provincial Center for Disease Control and Prevention, Shandong Provincial Key Laboratory of Infectious Disease Control and Prevention, Jinan, Shandong, China

³School of Public Health, Cheeloo College of Medicine, Shandong University, Jinan, China

⁴School of Public Health, Shandong Second Medical University, Weifang, Shandong, China

⁵School of Public Health, Shandong First Medical University and Shandong Academy of Medical Sciences, Jinan, Shandong, China

Received: 18 July 2024 / Accepted: 9 June 2025

Published online: 01 July 2025

References

- Bennett MJ, Kaasa S, Barke A, Korwisi B, Rief W, Treede RD. The IASP classification of chronic pain for ICD-11: chronic cancer-related pain. *Pain*. 2019;160(1):38–44. <https://doi.org/10.1097/j.pain.0000000000001363>.
- Snijders RA, Brom L, Theunissen M, van den Beuken-van Everdingen MH. Update on prevalence of pain in patients with cancer 2022: a systematic literature review and meta-analysis. *Cancers*. 2023;15(3):591. <https://doi.org/10.3390/cancers15030591>.
- Carlson LE, Waller A, Groff SL, Giese-Davis J, Bultz BD. What goes up does not always come down: patterns of distress, physical and psychosocial morbidity in people with cancer over a one year period. *Psycho-oncology*. 2013;22(1):168–76. <https://doi.org/10.1002/pon.2068>.
- Firkins J, Hansen L, Driessnack M, Dieckmann N. Quality of life in chronic cancer survivors: a meta-analysis. *J Cancer Surviv*. 2020;14(4):504–17. <https://doi.org/10.1007/s11764-020-00869-9>.
- Swarm RA, Paice JA, Anghelescu DL, Are M, Bruce JY, Buga S, et al. Adult Cancer pain, version 3.2019, NCCN clinical practice guidelines in oncology. *J Natl Compr Canc Netw*. 2019;17(8):977–1007. <https://doi.org/10.6004/jnccn.2019.0038>.
- Zhixiang C, Ke M, Jianxiu W, Dequan W, Cuiwei S, Jinfeng L, et al. A Chinese guideline for the diagnosis and treatment of chronic cancer-related pain (2024 edition). *Chin J Painol*. 2024;20(5):646–64. <https://doi.org/10.3760/cmaj.cn101658-20240815-00115>.
- Floriana I, Federico D. Quality of Life and Biopsychosocial Paradigm: A Narrative Review of the Concept and Specific Insights. In: Floriana I, Federico D, Simon George T, editors. *Quality of Life*. Rijeka: IntechOpen; 2020. p. Ch. 1. <https://doi.org/10.5772/intechopen.91877>.
- Luckett T, Davidson PM, Green A, Boyle F, Stubbs J, Lovell M. Assessment and management of adult cancer pain: a systematic review and synthesis of recent qualitative studies aimed at developing insights for managing barriers and optimizing facilitators within a comprehensive framework of patient care. *J Pain Symptom Manag*. 2013;46(2):229–53. <https://doi.org/10.1016/j.jpaininsymman.2012.07.021>.
- Quinten C, Coens C, Mauer M, Comte S, Sprangers MA, Cleeland C, et al. Baseline quality of life as a prognostic indicator of survival: a meta-analysis of individual patient data from EORTC clinical trials. *Lancet Oncol*. 2009;10(9):865–71. [https://doi.org/10.1016/s1470-2045\(09\)70200-1](https://doi.org/10.1016/s1470-2045(09)70200-1).
- Zaza C, Baine N. Cancer pain and psychosocial factors: A critical review of the literature. *J Pain Symptom Manag*. 2002;24(5):526–42. [https://doi.org/10.1016/s0885-3924\(02\)00497-9](https://doi.org/10.1016/s0885-3924(02)00497-9).
- Filippini C, Masiero M, Pizzoli SFM, Grasso R, Ferrucci R, Pravettoni G. A comprehensive analysis of the Cancer chronic pain experience: A narrative review. *Cancer Manage Res*. 2022;14:2173–84. <https://doi.org/10.2147/cmar.S355653>.
- Tavoli A, Montazeri A, Roshan R, Tavoli Z, Melyani M. Depression and quality of life in cancer patients with and without pain: the role of pain beliefs. *BMC Cancer*. 2008;8. <https://doi.org/10.1186/1471-2407-8-177>.
- Bedard G, Davies A, McDonald R, Hawley P, Buchanan A, Popovic M, et al. Breakthrough cancer pain: a comparison of surveys with European and Canadian patients. *Support Care Cancer*. 2015;23(3):791–6. <https://doi.org/10.1007/s00520-014-2426-6>.
- Durodié B, Wainwright D. Terrorism and post-traumatic stress disorder: a historical review. *Lancet Psychiatry*. 2019;6(1):61–71. [https://doi.org/10.1016/s2215-0366\(18\)30335-3](https://doi.org/10.1016/s2215-0366(18)30335-3).
- Engel GL. NEED FOR A NEW MEDICAL MODEL – CHALLENGE, FOR BIOMEDICAL SCIENCE. *Sci (New York NY)*. 1977;196(4286):129–36. <https://doi.org/10.1126/science.847460>.
- Gatchel RJ, Peng YB, Peters ML, Fuchs PN, Turk DC. The biopsychosocial approach to chronic pain: scientific advances and future directions. *Psychol Bull*. 2007;133(4):581–624. <https://doi.org/10.1037/0033-2909.133.4.581>.
- Yan W, Zhonglin W. Item parceling strategies in structural equation modeling. *Adv Psychol Sci*. 2011;19(12):1859–67. <https://doi.org/10.3724/SPJ.1042.2011.01859>.
- Kline RB. Principles and practice of structural equation modeling. Guilford; 2023. <https://journals.library.ualberta.ca/csp/index.php/csp/article/download/29418/21439>.
- Scholz JPPM. Finally, a systematic classification of pain (the ICD-11). 2019;19. <http://dx.doi.org/https://www.medcentral.com/pain/finally-systematic-classification-pain-icd-11>
- Cleeland CS, Ryan K. The brief pain inventory. *Pain Research Group*. 1991;20(20):143–7. <http://dx.doi.org/https://www.researchgate.net/profile/David-Rodriguez-Medina-2/post/Does-anybody-know-about-a-validating-pain-scale-for-Spanish-speakers/attachment/59d6400079197b807799c48d/AS%3A429160729452544%401479331299616/download/BPI.pdf>
- Kumar SPJL. Utilization of brief pain inventory as an assessment tool for pain in patients with cancer: a focused review. *Indian J Palliat Care*. 2011;17(2):108. <https://doi.org/10.4103/0973-1075.84531>.
- Wang XS, Mendoza TR, Gao S-Z, Cleeland CS. The Chinese version of the brief pain inventory (BPI-C): its development and use in a study of cancer pain. *Pain*. 1996;67(2):407–16. [https://doi.org/10.1016/0304-3959\(96\)03147-8](https://doi.org/10.1016/0304-3959(96)03147-8).
- Piccinin C, Kuliš D, Bottomley A, Bjordal K, Coens C, Darlington A-S, EORTC QUALITY OF LIFE GROUP ITEM LIBRARY USER GUIDELINES. [http://eprints.soton.ac.uk/id/eprint/467310](http://dx.doi.org/http://eprints.soton.ac.uk/id/eprint/467310)
- Cheng J-x, Liu B-l, Zhang X, Zhang Y-q, Lin W, Wang R et al. The validation of the standard Chinese version of the European Organization for Research and Treatment of Cancer Quality of Life Core Questionnaire 30 (EORTC QLQ-C30) in pre-operative patients with brain tumor in China. *BMC medical research methodology*. 2011;11:1–9. <http://dx.doi.org/http://www.biomedcentral.com/1471-2288/11/56>
- Bujang MA, Omar ED, Baharum NAJTM. A review on sample size determination for cronbach's alpha test: a simple guide for researchers. 2018;25(6):85. <https://doi.org/10.21315/mjms2018.25.6.9>
- Bonafé FSS, de Campos LA, Marôco J, Campos JADB. Brief pain inventory: a proposal to extend its clinical application. *Eur J Pain*. 2019;23(3):565–76. <https://doi.org/10.1002/ejp.1330>.
- Fornell C, Larcker DFJ. Evaluating structural equation models with unobservable variables and measurement error. 1981;18(1):39–50. <https://doi.org/10.2307/3151312>
- Bonafé FSS, de Campos LA, Marôco J, Campos J. Brief pain inventory: A proposal to extend its clinical application. *Eur J Pain*. 2019;23(3):565–76. <https://doi.org/10.1002/ejp.1330>.
- Farrar JT. Pain interference on quality of life is not just about physical function. *Pain Med*. 2019;20(2):211–2. https://doi.org/10.1093/pm/pny147_2.
- Karayannis NV, Sturgeon JA, Chih-Kao M, Cooley C, Mackey SC. Pain interference and physical function demonstrate poor longitudinal association in people living with pain: a PROMIS investigation. *Pain*. 2017;158(6):1063–8. <https://doi.org/10.1097/j.pain.0000000000000881>.
- Ovayolu N, Ovayolu Ö, Serçe S, Tuna D, Pirbudak Çöçelli L, Sevinç A. Pain and quality of life in Turkish cancer patients. *Nurs Health Sci*. 2013;15(4):437–43. <https://doi.org/10.1111/nhs.12047>. Epub 2013 Mar 11.
- Rodriguez C, Ji M, Wang H-L, Padhya T, McMillan SC. Cancer pain and quality of life. *J Hospice Palliat Nurs*. 2019;21(2):116–23. <https://doi.org/10.1097/njh.0000000000000507>.
- Pandelani FF, Nyalunga SLN, Mogotsi MM, Mkhathshwa VB. Chronic pain: its impact on the quality of life and gender. *Front Pain Res*. 2023;4. <https://doi.org/10.3389/fpain.2023.1253460>.
- Vu S, Pham P, Nguyen C, Trinh T. Pain and its interference with daily activities among patients with Cancer in Vinh Phuc Province and some related factors. *Int J Med Sci Clin Res Stud*. 2023;03. <https://doi.org/10.47191/ijmscrs/v3-i8-38>

35. Tegegn HG, Gebreyohannes EA. Cancer pain management and pain interference with daily functioning among Cancer patients in Gondar university hospital. *Pain Res Manag*. 2017;2017:5698640. <https://doi.org/10.1155/2017/5698640>.
36. Boveltdt NT, Vernooij-Dassen M, Burger N, Ijsseldijk M, Vissers K, Engels Y. Pain and its interference with daily activities in medical oncology outpatients. *Pain Physician*. 2013;16(4):379–89. <http://dx.doi.org/https://www.painphysicianjournal.com/current/pdf?article=MTk1MQ%3D%3D&journal=76>.
37. Zhu H, Lin R, Wang J, Ruan S, Hu T, Lei Y, et al. Cognitive function and its associated factors among patients with cancer pain: a multicentre cross-sectional study in China. *BMJ Open*. 2024;14(1):e071417. <https://doi.org/10.1136/bmjopen-2022-071417>.
38. Ko HJ, Seo SJ, Youn CH, Kim HM, Chung SE. The association between pain and depression, anxiety, and cognitive function among advanced cancer patients in the hospice ward. *Korean J Family Med*. 2013;34(5):347. <https://doi.org/10.4082/kjfm.2013.34.5.347>.
39. Li X-M, Xiao W-H, Yang P, Zhao H-X. Psychological distress and cancer pain: results from a controlled cross-sectional survey in China. *Sci Rep*. 2017;7(1):39397. <https://doi.org/10.1038/srep39397>.
40. Miettinen T, Kautiainen H, Mäntyselkä P, Linton SJ, Kalso E. Pain interference type and level guide the assessment process in chronic pain: categorizing pain patients entering tertiary pain treatment with the brief pain inventory. *PLoS ONE*. 2019;14(8):e0221437. <https://doi.org/10.1371/journal.pone.0221437>.
41. Mory L-N, de Oliveira Fernandes D, Mancini C, Mouthon M, Chabwine JN. The affective dimension of pain appears to be determinant within a pain–insomnia–anxiety pathological loop in fibromyalgia: a case-control study. *J Clin Med*. 2022;11(12):3296. <https://doi.org/10.3390/jcm11123296>.
42. Vadivelu N, Kai AM, Kodumudi G, Babayan K, Fontes M, Burg MM. Pain and psychology—a reciprocal relationship. *Ochsner J*. 2017;17(2):173–80. <http://dx.doi.org/https://pmc.ncbi.nlm.nih.gov/articles/PMC28638291/>.
43. Thoits PA. Mechanisms linking social ties and support to physical and mental health. *J Health Soc Behav*. 2011;52(2):145–61. <https://doi.org/10.1177/0022146510395592>.
44. Galloway SK, Meadors P, Boselli D, Walsh D. Anxiety, depression, pain, and social support in a large representative cancer population. *Journal of Clinical Oncology*. 2019;37(31_suppl):76–. https://doi.org/10.1200/JCO.2019.37.31_suppl.76.
45. Wiech K. Deconstructing the sensation of pain: the influence of cognitive processes on pain perception. *Sci (New York NY)*. 2016;354(6312):584–7. <http://doi.org/10.1126/science.aaf8934>.
46. Gross JJ. Emotion regulation: current status and future prospects. *Psychol Inq*. 2015;26(1):1–26. <https://doi.org/10.1080/1047840X.2014.940781>.
47. Kwekkeboom KL. CANCER SYMPTOM CLUSTER MANAGEMENT. *Semin Oncol Nurs*. 2016;32(4):373–82. <https://doi.org/10.1016/j.soncn.2016.08.004>.
48. Boccia ML. Social relationships and relational pain in brain tumor patients and their partners. *Frontiers in pain research*. (Lausanne Switzerland). 2022;3:979758. <https://doi.org/10.3389/fpain.2022.979758>.
49. Kroenke CH, Alexeeff S, Kushi LH, Kwan ML, Matthews KA. Clustering of social and physical pain variables and their association with mortality in two Population-Based cohorts. *Psychosom Med*. 2021;83(3):228–38. <https://doi.org/10.1097/psy.0000000000000910>.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.